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CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

BIOLOGY

Paper 03 – General Proficiency

1 hour

02 JUNE 2005 (a.m.)

In addition to the 1 hour, candidates are allowed a reading time of 10 minutes. Writing may begin during the 10-minute period.

This paper is divided into THREE sections. Candidates MUST answer ONE question from EACH section. Where appropriate, answers should be illustrated by diagrams.

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SECTION A

Answer ONE question from this section.

1. (a) With the aid of a diagram, describe the structure of the male reproductive system in humans and explain why this system is well-suited to its functions. **(8 marks)**

(b) (i) Explain the role of the male hormone (male gonadotropin) secreted by the gonads in controlling growth and development. **(2 marks)**

(ii) Give TWO reasons why a man who is undergoing a sex change operation is given a course of female gonadotropins. **(4 marks)**

(c) (i) Plant growth substances are often referred to as plant 'hormones'.

Do you consider this reference appropriate? Give TWO reasons for your answer. **(2 marks)**

(ii) Sometimes plant growth substances are sprayed onto plants to improve certain characteristics, for example, to hasten fruit ripening.

Suggest TWO **disadvantages** of this practice. **(4 marks)**

Total 20 marks

2. (a) With the aid of a diagram, describe the structure of a kidney tubule and explain why its structure makes it well-suited to its functions. **(8 marks)**

(b) Anti-diuretic hormone (ADH) causes changes in the functioning of the kidney tubule. Explain how these changes are brought about. **(4 marks)**

(c) (i) Suggest TWO reasons why osmo-regulation is important in plants. **(4 marks)**

(ii) A farmer selected crops which had deeply penetrating roots, and leaves with thick cuticles for a particular area of land he wished to cultivate.

Fully explain why a farmer would select these features for his crops. **(4 marks)**

Total 20 marks

SECTION B

Answer ONE question from this section.

3. (a) (i) Using an **annotated** diagram ONLY describe the structure of a dicotyledonous leaf as seen in cross section. **(4 marks)**

(ii) Describe TWO ways in which the leaf is well-suited for its functions. **(4 marks)**

(b) (i) The citrus swallowtail butterfly lays eggs on the leaves of citrus plants.

Fully explain what is the advantage to the butterfly of laying eggs on the leaves. **(4 marks)**

(ii) Although the citrus swallowtail larvae are parasitic on citrus plants, they do not necessarily kill the plants. Suggest why this is an **advantage to the butterfly**. **(2 marks)**

(c) (i) Crop yield may be affected by many factors including a covering of mould which forms a black layer on the upper surface of leaves.

Fully explain how a black layer covering the leaves might affect crop yield. **(4 marks)**

(ii) Suggest ONE reason why a lack of water may result in reduced crop yield. **(2 marks)**

Total 20 marks

4. (a) (i) Using an **annotated** diagram ONLY illustrate the sites where digestion occurs in humans. **(4 marks)**

(ii) Describe TWO features of the system which make it suitable for digestion. **(4 marks)**

(b) With reference to the products of digestion in humans, illustrate why the liver is sometimes referred to as the 'chemical laboratory' of the body. **(6 marks)**

(c) As producers, plants are very important in every ecosystem.

Apart from nutrition suggest THREE other ways in which plants add value to an ecosystem. **(6 marks)**

Total 20 marks

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SECTION C

Answer ONE question from this section.

5. (a) Outline the main features of mitosis and meiosis. **(8 marks)**

(b) (i) The study of genetics is based on our knowledge of the role and behaviour of chromosomes.

Identify TWO features of the role and behaviour of chromosomes that are important in predicting the outcome of crosses between gametes. **(4 marks)**

(ii) Two grey mice when mated, produced a total of 8 white, 9 black and 17 grey offspring. Student A suggested that grey coat colour was dominant in the mice. Student B strongly disagreed.

Which student is right? Explain your answer using appropriate symbols. **(8 marks)**

Total 20 marks

6. (a) (i) Outline the factors involved in the transmission and control of AIDS **and** a **named** vector-borne disease.

(ii) Suggest TWO reasons why it is important to determine the causes of disease. **(12 marks)**

(b) In terms of the spread of disease in plants, a vegetable garden can be considered a 'risky' environment compared to a forest.

Do you agree with this view? Support your answer with reasons. **(4 marks)**

(c) Concerted efforts are currently made by certain Caribbean governments to preserve forested areas although there is the need to provide housing for people.

Suggest TWO long-term biological benefits of a government's investment in forest conservation. **(4 marks)**

Total 20 marks

END OF TEST